

# WASP-43b

R-2538

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-43\_140403 · created 2026-08-25T23:53:06+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456750.70784 +/- 0.0006 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.183 +/- 0.011                  |
| Transit depth (Rp/Rs)^2                           | 3.37 +/- 0.4 [%]                 |
| Orbital Inclination (inc)                         | 81.99 +/- 0.76                   |
| Airmass coefficient 1 (a1)                        | 1.0037 +/- 0.0099                |
| Airmass coefficient 2 (a2)                        | -0.003 +/- 0.04                  |
| Scatter in the residuals of the lightcurve fit is | 0.98 %                           |
| Best Comparison Star                              | #2 - [47, 418]                   |
| Optimal Aperture                                  | 3.58                             |
| Optimal Annulus                                   | 8.17                             |
| Transit Duration (day)                            | 0.0521 +/- 0.0024                |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.183 ± 0.011 vs 0.1594 (deviation 2.13σ) |
| Duration fit vs published | 0.0521 d vs 0.0512 d (deviation 0.35σ)    |
| Mid-time O–C              | 1.5 min                                   |
| Depth SNR                 | 16.5                                      |
| Residual scatter          | 0.98 %                                    |

### Light curve

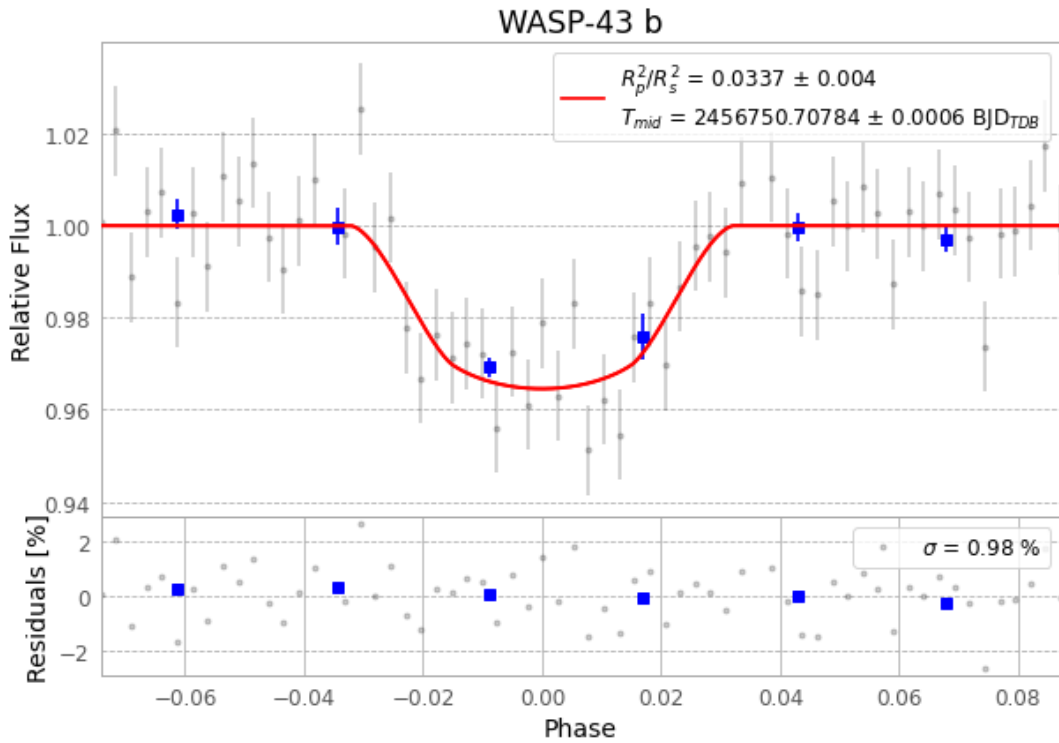


Figure 1 — FinalLightCurve\_WASP-43 b\_2014-04-02.png (SHA-256 f79ca021bfea120f...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [210, 241], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 cbf5820fa324242b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

64 FITS frames (42 MB), WASP-43140403032413.FITS → WASP-43140403063312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140403.log (26def7607698...), FinalLightCurve\_WASP-43 b\_2014-04-02.png (f79ca021bfea...), FinalLightCurve\_WASP-43 b\_2014-04-02.csv (3bfc0b21c867...)

## Compute resources

Wall time 257.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:53Z.